

Report "Distributed Robot Perception"

Hafsa Hamzaoui 256200* and Iuliia Sharipova 256295*

*University of Trento, Italy

Abstract—This report presents a fully distributed multi-robot system for the detection and localization of a gas leak in a constrained two-dimensional environment. Five homogeneous ground robots patrol the area using a Lloyd–Voronoi coverage strategy with active patrol inside each cell; once any robot detects the plume, the team fuses its observations through a Distributed Kalman Filter in information form, combined with weighted gossip consensus, to estimate the source location without a central coordinator. The system also handles sensor degradation through a trust-based consensus mechanism, and supports a separate formation-control mode driven by the same gossip machinery. We evaluate the system through five paired-seed experiments totalling 365 simulations. The team locates the source within tolerance (50 px) in all 100 baseline runs at full connectivity, continues to work correctly even when the communication graph is disconnected 100% of the time ($r_c = 200$ px), tolerates up to two simultaneously degraded sensors in line with the Pasqualetti–Bicchi–Bullo bound $k < n/2$, and shows no critical agent in a leave-one-out ablation across all five robot identities.

I. INTRODUCTION

Industrial gas leaks are colorless, often odorless, and their spread area is constantly changing. Fixed sensor networks have difficulty pinpointing the source of a leak, as the center of the plume shifts with the wind. A mobile robotic team performs active scanning and converges on the source as soon as any robot detects it, making this task a natural application of the theory of distributed estimation.

The reference work for this problem is the article "Cooperative UAVs Gas Monitoring using Distributed Consensus" [1]. Similar to their work, our project considers a team of autonomous agents that cooperatively explore an environment, detect a gas leak, and estimate the source location using only local communication.

However, our approach introduces several differences. Source localization is performed using a Distributed Kalman Filter (DKF) in information form [2], which allows the robots to continuously refine a shared estimate of the source position. Environment exploration is based on Lloyd–Voronoi coverage [3], ensuring that the robots distribute themselves efficiently over the area. In addition, the system accounts for sensor degradation through a trust-based consensus mechanism and can switch to a consensus-based formation control mode when required.

II. PROBLEM FORMULATION

We consider a team of $n = 5$ homogeneous ground robots, indexed by $i \in \{1, \dots, n\}$, tasked with localizing a gas leak in a planar environment bounded by $\Omega = [0, L_x] \times [0, L_y]$, where $L_x = 900$, $L_y = 600$ pixels (approximately 90×60 m

at 0.1 m/px). Each robot is treated as a point moving with first-order dynamics,

$$\dot{\xi}_i(t) = u_i(t), \quad \|u_i\| \leq v_{\max}, \quad (1)$$

where $\xi_i \in \mathbb{R}^2$ is its position and $v_{\max} = 2$ px/step limits its velocity. Each robot is equipped with a gas sensor with a detection range of $\rho = 70$ px, a communication radio with a range of $r_c = 500$ px.

Somewhere in Ω , at a time and location unknown to the robots, a gas source $p_s = (x_s, y_s)$ appears, which is resampled at the beginning of each new leak cycle. The system operates in three modes. In Modes 1 and 2 the team must agree on an estimate $\hat{p}_s$ satisfying $\|e_s\| < \rho_m = 50$ px without any central coordinator. In Mode 3 the robots maintain a prescribed geometric formation while navigating the environment. Essentially, this must happen without any central coordinator: each robot can only execute its own local copy of the algorithm, using its own measurements and whatever its neighbors within radius r_c choose to share with it.

III. ADOPTED MODELS

A. Communication System

Since robots can communicate only with those in their vicinity, it is appropriate to describe their connectivity as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with one vertex for each robot and an edge $(i, j) \in \mathcal{E}$ whenever $\|\xi_i - \xi_j\| \leq r_c$. From this graph, we obtain the usual adjacency matrix A , the degree matrix D , and the Laplacian $L = D - A$.

What matters most for the purposes of our study is the second smallest eigenvalue of L , the *Fiedler value* $\lambda_2(L)$, also known as the *algebraic connectivity* of the network: $\lambda_2 = 0$ exactly when the graph is disconnected, and the larger λ_2 is, the faster information spreads through the team.

In fact, λ_2 directly determines the speed at which the gossip algorithm converges:

$$\|\eta(k+1)\| \leq |1 - \varepsilon\lambda_2| \|\eta(k)\|, \quad (2)$$

for a step size $\varepsilon \in (0, 2/\lambda_{\max})$, where $\eta(k)$ denotes the disagreement vector at iteration k [4]. Equivalently, in the discrete-time stochastic-matrix formulation of linear consensus, $x(k+1) = Qx(k)$, the convergence rate is governed by the *essential spectral radius* $\rho_2(Q)$ (the second-largest eigenvalue of Q). For a symmetric gossip mixing matrix $Q = I - \varepsilon L$ the two viewpoints are equivalent: $\rho_2(Q) = |1 - \varepsilon\lambda_2|$ [5]. Because of this central role, λ_2 is calculated analytically at every step of the simulation and displayed in real time as an indicator of the team's current degree of cohesion.

B. System Model

1) *Gas plume*: The simulator offers *two interchangeable models* for the gas plume, selectable via a single configuration selector.

a) *Pasquill–Gifford analytical model*: Assuming, as is customary, that the two random variables modelling the plume spread are independent, the joint pdf factorises as $p_{y,z}(y, z) = p_y(y) p_z(z)$, so that the concentration is:

$$C(x, y, z, H_s) = \frac{Q}{w} \frac{1}{2\pi\sigma_y(x)\sigma_z(x)} e^{-y^2/(2\sigma_y^2(x))} \times \left[e^{-(z-H_s)^2/(2\sigma_z^2(x))} + e^{-(z+H_s)^2/(2\sigma_z^2(x))} \right], \quad (3)$$

which accounts for the ground reflection effect, where Q is the source mass-emission rate and w is the wind speed [1].

b) *Diffusion approximation*: A more lightweight alternative is the advection–diffusion PDE:

$$\frac{\partial C}{\partial t} = D_c \nabla^2 C - \mathbf{w} \cdot \nabla C - \kappa C + Q \delta(x - x_s, y - y_s), \quad (4)$$

with $D_c = 0.45$, $\kappa = 0.003$, and a wind field $\mathbf{w}$ that shifts slowly over time. Despite its simplicity, this PDE reproduces the same Gaussian steady-state profile as (3) at a fraction of the computational cost.

2) *Sensor model*: For the gas sensor, we adopt the model of the MiniPID 2 photoionisation detector (Ion Science), the same device used in the reference work [1].

Assuming a gas source positioned at $\mathbf{p}_s = [x_s, y_s, H_s]^\top$ and the i -th robot located at ξ_i , with the sensor co-located at the robots centre of mass (without loss of generality), the gas concentration at the sensor is a modified version of (3):

$$C_i(\xi_i, \mathbf{p}_s) = \frac{Q}{w} \frac{1}{2\pi\sigma_y(x_i - x_s)\sigma_z(x_i - x_s)} \times e^{-(y_i - y_s)^2/(2\sigma_y^2(x_i - x_s))} \times \left[e^{-(z_i - H_s)^2/(2\sigma_z^2(x_i - x_s))} + e^{-(z_i + H_s)^2/(2\sigma_z^2(x_i - x_s))} \right]. \quad (5)$$

Accounting for the sensor’s limited range, the measurement is

$$h_i(\xi_i) = \text{sat}(C_i(\xi_i, \mathbf{p}_s)), \quad (6)$$

where $\text{sat}(\cdot)$ clips the reading at the upper end of the sensor’s calibration range. In the implementation, gas concentration is represented in normalised units $[0, 1]$, with 1 corresponding to approximately the maximum reading of the MiniPID 2 datasheet (~ 100 ppb). The residual sensor uncertainty about $h_i(\xi_i)$ is modelled as a zero-mean Gaussian random effect whose variance has been characterised empirically from the sensor datasheet, satisfying the zero-mean assumption $\mathbb{E}\{\varepsilon\} = 0$ which the Kalman filter assumes for an unbiased estimate.

3) *Source model*: Finally, the source $\mathbf{p}_s$ itself is considered fixed once it appears, but unknown to the robots, and is uniformly resampled within Ω at the beginning of each new leak cycle.

IV. SOLUTION

Throughout this work, robots are modeled using first-order kinematics. This simplification allows us to focus on the behavior and convergence of the distributed consensus and coverage algorithms without introducing additional complexity from the robot dynamics.

The framework integrates four distributed algorithms that run locally on each robot. Each agent operates autonomously without global synchronization, ensuring scalability and robustness.

A. Lloyd–Voronoi Coverage with Active Patrol

To achieve optimal spatial coverage, we minimize the cost function

$$H(p) = \sum_{i=1}^n \int_{V_i} \phi(q) \|q - p_i\|^2 dq, \quad (7)$$

where V_i is the Voronoi cell of robot i and $\phi(q) \equiv 1$ for uniform coverage. In the classical Lloyd algorithm, the system converges to a Centroidal Voronoi Tessellation (CVT), where each robot remains at the centroid of its assigned cell. However, such a static configuration is not ideal for hazard localization. If a leak occurs in a region far from the cell centroid, it may remain undetected for a prolonged period due to the robots’ limited sensing range.

To address this we introduce active patrol within cells. Rather than staying at the centroid, each robot treats its Voronoi cell as its area of responsibility and continuously moves between randomly generated waypoints inside the cell. Once a waypoint is reached, a new target is sampled. This allows robots to explore their entire assigned regions instead of remaining stationary. As a result, the probability of detecting a hazard anywhere in the environment increases over time while maintaining the collision-free spatial partitioning provided by the Voronoi tessellation.

B. Random Walk Exploration (Benchmark Strategy)

To compare the proposed Lloyd–Voronoi patrol approach with the exploration strategies presented in the reference work, we also implement the *Random Walk* strategy proposed in [1].

This strategy is not available as an interactive simulation mode; instead, it is implemented as a configurable option (EXPLORATION_STRATEGY in `config/default.py`).

In this approach, each robot moves at a constant speed while its heading (θ_i) is perturbed at every time step by a uniformly distributed random angular increment:

$$\theta_i(t) = \theta_i(t - \Delta t) + \nu, \quad \nu \sim \mathcal{U}(-\theta_M, \theta_M), \quad (8)$$

with $\theta_M = 10^\circ$.

When a robot reaches the boundary of the environment, its heading is reset to the inward normal direction θ_{in} perturbed by a random offset bounded by $\theta_r = 75^\circ$:

$$\theta_i(t) = \theta_{\text{in}} + \nu_r \theta_r, \quad \nu_r \sim \mathcal{U}(-1, 1), \quad (9)$$

where θ_{in} denotes the inward normal direction at the boundary.

To prevent collisions, a pairwise repulsion mechanism is activated whenever the distance between two robots satisfies ($\|\xi_i - \xi_j\| < d_m$).

C. Weighted Gossip Consensus

A first communication channel runs at the level of the robots' *position estimates*: each communication round randomly selects a pair (i, j) of neighbours and merges their estimates of each other's positions [5]:

$$\hat{x}_i^+ = \hat{x}_j^+ = \frac{w_i \hat{x}_i + w_j \hat{x}_j}{w_i + w_j}, \quad (10)$$

with weight $w_k = q_k$, where $q_k \in \{1.0, 0.55, 0.18, 0\}$ is a discount factor assigned by the degradation stage of robot k (healthy, noisy, failing, dead) — an empirical reliability score rather than the inverse-variance precision $1/\sigma_k^2$ that would be LS-optimal under Ch. 17.2.2 of the course notes. Eighteen rounds are executed per simulation step. The second, more important, fusion channel acts on the DKF beliefs themselves; it follows the same pairwise structure but uses additive information-form updates and an enriched weight $w_k = q_k \cdot \tau_k$ that also incorporates reputation (see §IV-D and §IV-F).

D. Distributed Kalman Filter

Among the standard hierarchy of Bayesian estimators (LS, WLS, MMSE, MAP, ML), the Kalman filter is the sequential MMSE estimator for linear-Gaussian models. Our setting fits this assumption: the source p_s is static once the leak appears, sensor noise is approximately zero-mean Gaussian, and a robot's belief is well represented by a Gaussian. To track the location of the leak, each robot therefore maintains a Gaussian distribution $\mathcal{N}(\mu_i, P_i)$ over the source coordinates p_s . Since the robots are constantly moving, their effective uncertainty grows between measurement updates; we account for this with the standard Kalman prediction step that adds a process-noise covariance

$$P_i \leftarrow P_i + Q, \quad Q = qI, \quad q = 25 \text{ px}^2,$$

which inflates the covariance by a fixed amount per step regardless of its current magnitude. The value $q = 25 \text{ px}^2$ was calibrated empirically to match the typical sensor-noise covariance in near-source observations.

When a robot first detects gas, it initialises its probabilistic estimate using a local observation: $\mu_i \leftarrow \text{peak}_i$, where peak_i is the location of the strongest sensor reading observed by robot i so far. Thereafter, at every step in which its sensor sees the plume, a healthy robot incorporates the current peak observation z_i through the standard *update step* of the filter in information form (Ch. 18.2 of the course notes),

$$\Omega_i \leftarrow \Omega_i + H^\top R^{-1} H, \quad y_i \leftarrow y_i + H^\top R^{-1} z_i, \quad (11)$$

with $H = I$ (the peak location observes the source position directly) and $R = \max(R_{\min}, R_0 (d/\rho)^2) I$: the measurement noise grows quadratically with the distance d between the robot and the observed peak ($R_{\min} = 100 \text{ px}^2$; R_0 is the

calibrated sensor-noise covariance), reflecting the empirically characterised, distance-scaled noise model of the sensor. Prediction and update together realise the two phases of the Kalman filter for our static-source case: uncertainty grows while a robot moves without seeing the plume, and contracts as it accumulates measurements from ever-closer, less noisy vantage points. Degraded robots do not perform this update — their corrupted measurement stream is precisely the fault modelled in Mode 2. For multi-robot fusion we work in the *information form* of the filter [2]: each Gaussian belief is reparameterised through the information matrix $\Omega_i = P_i^{-1}$ and the information vector $y_i = \Omega_i \mu_i$, so that pairwise integration becomes simple and additive:

$$\Omega_f = w_i \Omega_i + w_j \Omega_j, \quad (12)$$

$$y_f = w_i y_i + w_j y_j, \quad (13)$$

with $\mu_f = \Omega_f^{-1} y_f$. This is the time-varying counterpart of the distributed weighted least-squares fusion rule $F_n = \sum_i F_i$, $a_n = \sum_i a_i$ for static parameters (Ch. 18 notes). The additivity makes the scheme genuinely distributed and provides a principled alternative to the spiral heuristic of [1].

However, data exchange is not enough on its own to achieve reliable consensus. To prevent the network from jumping to an inaccurate conclusion, we apply a strict two-level confirmation criterion.

First, consensus is gated by a two-part observer requirement: at least two *healthy* observers (`has_detected` = True and not flagged as degraded) must have made their own independent observations, and the *total* number of detecting robots must satisfy

$$n_{\text{obs}} \geq \max(2, \lceil n_{\text{alive}}/2 \rceil) \quad (14)$$

The floor of two observers prevents a single agent from unilaterally declaring consensus in heavily depleted teams ($n_{\text{alive}} \leq 3$), which would violate the $k < n/2$ tolerance bound of Pasqualetti, Bicchi and Bullo [6] at the boundary.

Second, we track the degree of disagreement when consensus is reached. A design choice matters here: after 18 gossip rounds per step the fused estimates μ_i of connected robots coincide almost immediately, so their spread would read ≈ 0 from the very first fusion round and would certify nothing — a single biased robot could drag the merged value with it and the “agreement” would still look perfect. We therefore measure the spread of the *seed* beliefs μ_i^{seed} , each robot's own independent observation, which fusion never overwrites. Let $\mathcal{I}$ denote the set of trusted robots with an own observation, i.e. $\mathcal{I} = \{i : \text{self_observed}_i = \text{True} \wedge \bar{\tau}_i \geq \tau_{\text{exc}}\}$

$$\sigma(k) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \|\mu_i - \bar{\mu}\| \quad (15)$$

Agreement of independent observations is a stronger certificate than agreement of already-merged estimates.

We consider a location to be confirmed only if $\sigma < \Theta = 25$ pixels for 30 consecutive steps. This double-check serves as

our insurance from a common type of failure in distributed systems, where a single robot with a systematic error unintentionally “misleads” the rest of the team, leading to a false consensus before they have enough data of their own.

Convergence behaviour. For a static communication topology, the consensus error exponentially decays at a rate of $|1 - \varepsilon \lambda_2|$ (Equation 2). In our randomized setting, the topology changes as the robots move, but the average value of λ_2 , averaged over the window, remains limited and nonzero in all observed runs; therefore, the property of convergence is still preserved on average.

E. Scout Alert Mechanism

When the system switches to the LEAK DETECTED phase, the robots continue patrolling according to Lloyd’s algorithm, awaiting confirmation of a gas leak from at least 50% of the team members. If only one robot detects gas while the others are patrolling their cells at a significant distance, the leak may remain unconfirmed for a long time.

To speed up confirmation, we introduce a scout notification mechanism. Robot i is designated a *scout* as soon as it sets `has_detected=True`, which means it has detected a real anomaly within the range of its sensors. Every other robot j located within a communication radius r_c of the scout interrupts its Lloyd’s algorithm patrol and sets the scout’s position ξ_i as its target:

$$\xi_j^* = \xi_i \quad \text{if} \quad \|\xi_i - \xi_j\| \leq r_c \text{ and } \text{has_detected}_i. \quad (16)$$

It is important to note that j approaches the *location* of i , not the *estimate* of i (which may be biased, especially if i is a robot with impaired functionality). As soon as j is near the gas, its own sensor generates a clean initial DKF signal.

The scout robot with degraded performance demonstrates additional behavior: instead of moving toward its own (distorted) DKF estimate, it remains at its current location, acting as a stationary beacon for the others. This prevents a situation in which the robot with degraded performance leads the team away from the correct path with its incorrect estimate of the source’s location, while allowing it to play a useful role in reaching consensus.

F. Trust-Weighted Consensus

Inspired by the work of [6], each robot i maintains a trust estimate $\tau_{ij} \in [\tau_{\min}, \tau_{\max}]$ for each neighbor j , updated according to the EMA algorithm:

$$\tau_{ij} \leftarrow (1 - \alpha) \tau_{ij} + \alpha e^{-d_{ij}/d_\theta}, \quad (17)$$

where d_{ij} is the larger of two disagreement signals (DKF belief mismatch $\|\mu_i - \mu_j\|$ when both robots hold a belief, and the mismatch between i ’s position-estimate of j and j ’s actual position), $\alpha = 0.05$ is the EMA learning rate, and $d_\theta = 60$ px is the disagreement scale parameter.

The trust score τ_{ij} is multiplied into the DKF fusion weights in equations (12)–(13), which automatically reduces the contribution of untrustworthy agents during belief fusion.

Robots whose mean incoming reputation $\bar{\tau}_j$ falls below $\tau_{\text{exc}} = 0.4$ are additionally excluded from the gossip information fusion rounds and from the final calculation of `network_belief`; therefore, their drifting belief is displayed on the visualizer as a diverging marker $\times$, but does not disrupt the team’s consensus.

G. Formation Control

A consensus-based formation controller [7] assigns each robot an offset δ_i relative to the formation center c and directs it toward $\xi_i^* = c + \delta_i$.

Four configurations are implemented: **circle**: $\delta_i = \rho_f(\cos \theta_i, \sin \theta_i)$, $\theta_i = 2\pi i/n$, where c rotates around its axis; **line**: slots lying on a single line with an interval of s_f ; **V**: “bird flock” formation, with the vertex at slot 0; **triangle**: robots distributed along the three sides of an equal-sided triangle inscribed in a circle of radius ρ_f . If a robot fails, the slots are redistributed, and the formation is restored automatically.

V. IMPLEMENTATION DETAILS

The simulator is written in Python 3.11 with NumPy, SciPy, Shapely and Matplotlib. Architecture is strictly modular so any algorithm class can be swapped without touching the rest:

Folder	Contents
core/	Gossip, DKF, Lloyd, λ_2 , trust, formation
agents/	Robot class (state, sense, move)
environment/	Gas field, map, free space
scenario/modes/	The three mode state machines
simulation/	Orchestrator and event log
visualisation/	Matplotlib UI
experiments/	Batch runner and plotting scripts

A. State Machine and Interactive Modes

The simulator provides a three-mode interactive interface that allows users to observe the system in real time. **Mode 1** (Threat Localization) demonstrates the main distributed pipeline with all five robots healthy. **Mode 2** (Sensor Degradation) is a stress-testing mode in which malfunctioning agents are introduced; it exercises trust-weighted gossip, the scout mechanism, and the automatic exclusion of biased beliefs. **Mode 3** (Formation Control) demonstrates the consensus-based geometric controller, with shapes switchable via the C, N, V, T keys. The exploration strategy (Lloyd–Voronoi vs. random walk) is set globally in `config/default.py` since it is meant for repeatable autonomous testing.

In Modes 1–2 the system autonomously cycles through four internal phases: MONITOR (patrolling), LEAK DETECTED (scout-led investigation), REACHING CONSENSUS (DKF fusion), and REPORTED (source confirmation). Crucially, agents are never given global information: each robot must detect threats with its own noisy sensor, which gives a realistic assessment of distributed localization performance.

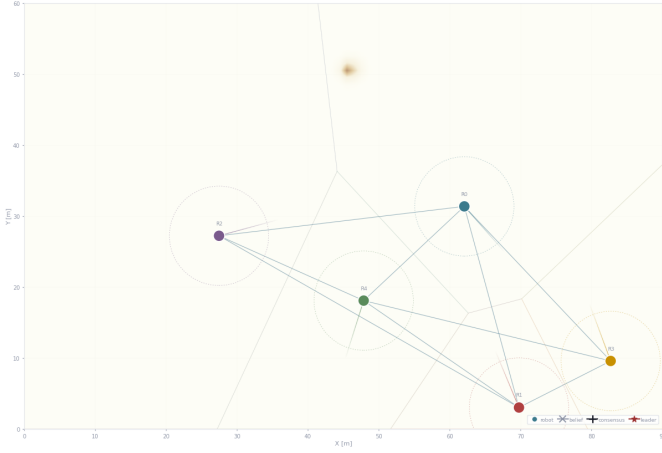


Fig. 1: Leak Detected phase. Robots continue their patrol until they detect the gas leakage.

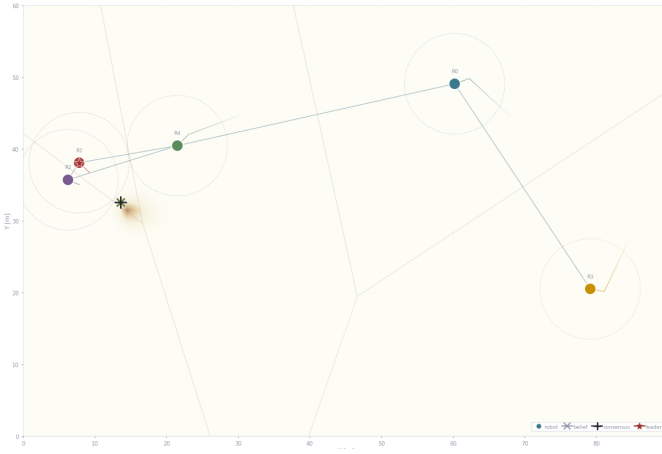


Fig. 2: Reaching Consensus phase. Agents that have detected gas maintain individual DKF beliefs ($\times$), while the network-wide consensus estimate ($+$) converges toward the true source location through gossip-based fusion.

B. Logging and Visualization

Each run produces a human-readable `.log` file (phase transitions, detections, consensus metrics) and a step-level `.csv` used for statistical analysis. Critical events also surface in the on-screen event log for real-time demonstrations.

The visualizer (Figs. 1, 2), inspired by [1], shows each robot as a circle with a dotted sensing ring and a fading trajectory trail. A red star marks the current leader (agent closest to the consensus mean); when the phase reaches REPORTED, an annotation arrow points to the agreed source $\hat{p}_s$. Gas concentration is rendered as a cream-to-brown gradient. Playback speeds run from $1\times$ to $16\times$ real time.

VI. EXPERIMENTAL RESULTS

A. What we wanted to measure, and why

Distributed perception is a system claim more than an algorithm claim: a result of the form “our team localizes the source within X pixels” is only meaningful if we can also say

how it behaves under *network* stress (loss of connectivity), *sensor* stress (degraded agents), *strategy* choice (Lloyd vs. Random Walk), and *redundancy* (what happens if any one robot is removed).

The five experiments below are organised around exactly these four axes, plus a baseline accuracy run that anchors the comparison.

B. Measurement protocol

All five experiments share one runner (experiments/runner.py) so that every CSV column has the same definition across experiments. The unified runner records four anchors per leak cycle:

- the step at which the gas first appears,
- the step at which *any* robot’s own sensor first crosses the detection threshold,
- the step at which $\geq 50\%$ of alive robots have detected (entry into the consensus phase),
- the step at which the consensus stability check finally triggers REPORTED.

From these we derive three non-overlapping time intervals: the *patrol* time before any robot detects the plume, the *investigation* time during which the scout mechanism recruits additional detectors, and the *fusion* time during which DKF and gossip lock in a consensus — plus the *total* time, which by construction is the sum of the three. Splitting the total this way matters because the three components are dominated by different mechanisms and have very different variances; aggregating them would hide the contribution of each algorithmic stage.

For every cycle we also store the *signed* error vector $\mathbf{e} = \hat{p}_s - p_s$, not only its norm. This enables the wind-bias analysis reported below.

a) Statistical reporting: All summary statistics are given as mean $\pm$ standard deviation across N independent runs. Where a parameter is being varied across conditions (Exp. 2–5), the *same* random seeds are replayed across conditions so that variance from source placement and initial robot positions cancels out. Sample sizes are chosen separately per experiment, with the largest budget ($N = 100$) going to the baseline accuracy measurement.

C. Experiment 1: baseline accuracy ($N = 100$)

This experiment characterises the system under conditions favourable to the perception pipeline: $r_c = 500$ px so the graph is near-fully-connected ($\bar{\lambda}_2 \approx 2.0$), all five robots healthy, Lloyd–Voronoi exploration. The goal is to isolate the contribution of the gossip/DKF stack from communication-topology and sensor-fault confounders. Table I reports the headline numbers.

All 100 runs complete inside the tolerance circle $\rho_m = 50$ px, and 99 of them inside the tighter 25-px circle — the continuous update step keeps refining every belief with close-range, low-noise observations, so the residual error is set by sensor noise near the source rather than by the first, distant seed. The fusion time of 31.9 ± 4.0 steps is worth

TABLE I: Mode-1 accuracy at $r_c = 500$ px, $N = 100$ runs.

Metric	mean $\pm$ std
Localization error $\ e\ $ [px]	7.8 ± 4.9
Patrol time [steps]	108.5 ± 125
Investigation time [steps]	122.7 ± 50
Fusion time [steps]	31.9 ± 4.0
Total time [steps]	263.0 ± 136
Mean λ_2 over cycle	2.0 ± 1.1
Within $\rho_m = 50$ px	100/100
Within 25 px	99/100

unpacking: the stability check in our implementation requires the consensus disagreement to stay below Θ for 30 consecutive steps before REPORTED is triggered, so the empirical fusion time sits essentially at that lower bound. Gossip and DKF therefore converge almost immediately once 50% of robots have detected — the contraction-rate prediction of (2) for $\bar{\lambda}_2 \approx 2$ is ≈ 11 steps to reduce disagreement by an order of magnitude, well below the safety window.

The other three intervals: patrol (108.5 steps), investigation (122.7 steps), total (263 steps) — show that the dominant cost is *finding the leak*, not fusing beliefs. Patrol time also has the largest variance ($\sigma \approx 124$ steps), which is expected: the distance from a randomly placed source to the nearest robot patrolling its Voronoi cell is high-variance by design.

a) *Downwind bias of the error vector*: The scatter of the signed error vector (e_x, e_y) over all 100 runs is shown in Figure 3; it is visibly skewed toward positive e_x . Component-wise means are

$$\bar{e}_x = 4.0 \pm 6.3 \text{ px}, \quad \bar{e}_y = 1.2 \pm 5.3 \text{ px},$$

both noticeably shifted away from zero. The wind vector in our diffusion plume points approximately $(+0.06, +0.02)$, mostly along $+x$ with a small $+y$ component, so the bias of the agreed estimate aligns with the wind direction. This is the standard behaviour of plume-following passive localization [8]: gas concentration peaks downwind of the actual source, and a robot that initialises its belief from the peak sensor reading inherits that downwind shift. The bias is small relative to the tolerance (~ 4 px vs. 50 px); repeated updates from ever-closer vantage points roughly halve the shift a seed-only initialisation would inherit, but cannot remove it entirely, since the concentration maximum itself sits downwind of the source. It does not threaten the localization guarantee, but is worth quantifying because it is a property of the physics rather than of the algorithm.

D. Experiment 2: how much communication is enough?

This experiment adapts the limited-communication modelling technique of Ch. 19.2.6 of the course (very large) notes: the Laplacian is masked element-wise by an indicator of the communication graph, so that agents outside r_c contribute zero coupling. The course notes apply this to formation keeping and measure RMSE of a formation-shape index against the ratio $r_c/L_{\text{formation}}$; we instead apply it to our localization pipeline and measure localization error and total time against r_c for the same parametric sweep idea.

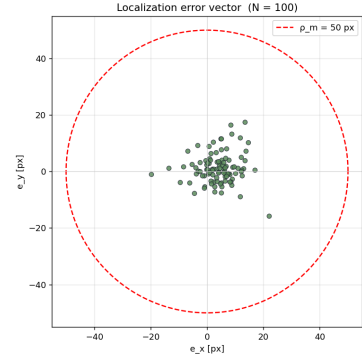


Fig. 3: Signed error vector (e_x, e_y) for all 100 runs at $r_c = 500$ px. Dashed circle: $\rho_m = 50$ px. The cloud is shifted toward $+x$, consistent with the wind direction in the plume model.

The same 15 random seeds were replayed under each of five values of the communication radius, $r_c \in \{200, 250, 300, 350, 500\}$ px. Replaying the *same* seeds across conditions cancels out the variance from where the leak happens to appear, so what remains can be attributed to the parameter we are sweeping. Table II shows the results.

TABLE II: Communication-radius sweep, paired seeds, $N = 15$ per radius.

r_c	$\ e\ $ [px]	total time [steps]	$\bar{\lambda}_2$	disconn. %
200	9.5 ± 5.7	1101 ± 787	0.000	100.0
250	8.9 ± 6.5	655 ± 411	0.006 ± 0.014	99.0
300	8.6 ± 2.7	300 ± 118	0.041 ± 0.069	95.0
350	8.1 ± 4.6	243 ± 139	0.528 ± 0.899	67.4
500	7.3 ± 3.9	242 ± 137	2.189 ± 1.079	6.4

Three findings are worth highlighting.

a) *The system works even when the graph is never connected*: At $r_c = 200$ px the communication graph is disconnected ($\lambda_2 = 0$) in 100% of all steps over all 15 runs, and at $r_c = 250$ px in 99%. Yet the completion rate is 15/15 at both radii, with localization errors of 9.5 and 8.9 px on average — well inside the tolerance. This is direct evidence that the scout-alert mechanism (§IV-E) is doing useful work: agents form transient *geographic* clusters around detected gas plumes, and gossip converges *locally* inside those clusters without needing a global path in the graph. Classical consensus theory [4], [7] requires connectivity for convergence to a single value; here, physical co-location substitutes for it when robots can move toward the phenomenon they are estimating.

b) *Connectivity buys speed more than it buys accuracy*: Going from $r_c = 200$ to $r_c = 500$ reduces the total time by a factor of $4.5\times$ ($1101 \rightarrow 242$ steps) but changes the localization error only marginally ($9.5 \rightarrow 7.3$ px — the accuracy curve is essentially flat across the sweep). The connection between connectivity and accuracy is therefore much weaker than between connectivity and convergence speed. The interpretation is that once a robot physically reaches the source, its own information-form update gives a high-quality belief regardless of whether it can share it with the rest of the team

— connectivity matters for *disseminating* the belief, not for *forming* it.

c) *Diminishing returns above $r_c \approx 350$ px*: The total time at $r_c = 350$ and $r_c = 500$ is essentially identical (243 vs. 242 steps). Beyond the point where the graph is connected most of the time, increasing r_c further does not buy additional speed — the bottleneck shifts to physical motion. In practical terms, there is little to gain from extending the radio range past the regime where $\bar{\lambda}_2$ is comfortably positive.

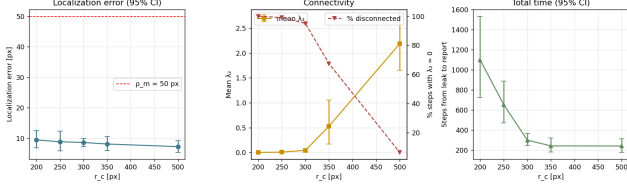


Fig. 4: Communication-radius sweep: localization error, mean λ_2 and disconnected fraction, total time. Error bars show $\pm$ standard deviation across $N = 15$ paired runs per radius.

E. Experiment 3: does the exploration strategy matter?

We ran $N = 30$ seeds under both Lloyd–Voronoi and Random Walk exploration with identical conditions otherwise, again replaying the same seeds across the two strategies so the contrast is paired. Our prior expectation was that Lloyd would be visibly faster because its partition avoids re-coverage, as the reference work [1] also reports.

TABLE III: Lloyd–Voronoi vs. Random Walk, paired seeds, $r_c = 500$, $N = 30$.

	Lloyd–Voronoi	Random Walk
Completion	30/30	30/30
$\ e\ $ [px]	9.3 ± 5.6	10.0 ± 6.7
Patrol time [steps]	112.3 ± 139	160.9 ± 222
Total time [steps]	267.3 ± 152	297.6 ± 243

The point estimates favour Lloyd on every metric ($\approx 30\%$ shorter patrol, slightly lower error), but the standard deviations show this is well within the run-to-run variability. The mean paired difference in error is -0.6 px, with a standard deviation of ≈ 8 px across the 30 paired differences — the spread is too large relative to the effect to conclude that Lloyd is reliably better at this sample size.

The paired design did not absorb as much variance as expected, because the strategy-by-trajectory noise is large: whether Lloyd or Random Walk wins on a given seed is not strongly determined by the seed itself. A point-estimate ranking favouring Lloyd is consistent with the literature [1], but a larger run would be needed to confirm it. What does hold at $N = 30$ is that both strategies complete 30/30 within tolerance, so the choice between them is one of preference rather than capability.

F. Experiment 4: sensor degradation (Mode 2)

Mode 2 is the fault-tolerance test. We compared three conditions on ten paired seeds: all robots healthy (*baseline*),

one robot forced into degradation at step 80 ($k=1$), and two robots forced into degradation ($k=2$). The radius was set to $r_c = 350$ px — a regime where the graph is disconnected 67% of the time, so trust-based exclusion has to do real work and is not trivially redundant with full connectivity.

TABLE IV: Mode-2 sensor degradation, paired seeds, $r_c = 350$, $N = 10$ per arm.

Condition	done	$\ e\ $ [px]	within ρ_m	total [steps]
baseline	10/10	6.1 ± 1.5	10/10	241 ± 105
$k=1$ deg.	10/10	6.9 ± 5.4	10/10	256 ± 99
$k=2$ deg.	10/10	10.8 ± 11.7	10/10	386 ± 189
$k=3$ deg.	8/10	10.6 ± 6.2	8/8	485 ± 245

All runs at $k \leq 2$ complete within the tolerance circle. Mean error roughly doubles between baseline and $k = 2$ ($6.1 \rightarrow 10.8$ px), and total time scales similarly ($241 \rightarrow 386$ steps), but neither exceeds the operating limit. This is consistent with the theoretical result of Pasqualetti, Bicchi and Bullo [6], which gives a tolerance bound of $k < n/2$ non-colluding faulty agents for a network of n . With $n = 5$ this means $k \leq 2$ is tolerable, $k = 3$ is not - consistent with the $k=3$ row, where 2 of 10 runs fail to reach consensus. The $k = 2$ runs all completing successfully suggests that the trust mechanism (§IV-F) together with the scout-as-beacon behaviour operate close to this theoretical bound.

The graceful, monotone increase in error with k is itself informative: a fault-tolerant system should degrade smoothly rather than collapse abruptly. The means at baseline and $k = 2$ (6.1 vs. 10.8 px) differ by several baseline standard deviations (1.5 px); the $k = 2$ spread is dominated by a heavier tail (std 11.7 px), while every single run stays within tolerance, so the degradation effect is visible yet bounded even at $N = 10$.

G. Experiment 5: is any single robot critical?

The previous experiments establish that the system works under various perturbations, but they do not on their own establish that the perception is *actually distributed*: a system can rely heavily on one “lead” agent and still appear robust to network or sensor stress, provided that agent is not the one being disturbed. To check for hidden centralisation we ran an ablation: 15 paired seeds, six conditions, each removing one of the five robots at startup (plus the baseline with all five present). If any single robot were critical, removing that robot would cause a visible spike in error or completion failure. Leave-one-out evaluation of this kind is standard practice for arguing that no hidden dependency on a single component exists [9].

Completion is 15/15 in every condition. The mean error across the five drop-conditions ranges from 8.4 px (drop R_3) to 10.9 px (drop R_4) — a spread of 2.5 px, smaller than the baseline’s own standard deviation (3.8 px). In other words, removing any one robot increases the error by an amount that is *within the natural run-to-run noise of the baseline itself*. There is no critical agent.

TABLE V: Per-robot ablation, paired seeds, $r_c = 500$, $N = 15$ per arm.

Condition	done	$\ e\ $ [px]	within ρ_m	total [steps]
baseline	15/15	7.5 ± 3.8	15/15	243 ± 84
drop R_0	15/15	8.8 ± 6.8	15/15	257 ± 159
drop R_1	15/15	8.7 ± 4.5	15/15	268 ± 159
drop R_2	15/15	10.2 ± 7.8	15/15	212 ± 106
drop R_3	15/15	8.4 ± 6.7	15/15	278 ± 167
drop R_4	15/15	10.9 ± 5.9	15/15	258 ± 161

All 90 runs, in every condition, land inside the tolerance circle. Some robots contribute slightly more than others (drop R_2 and drop R_4 show the largest mean errors), but losing any one of them never breaks the system — consistent with a genuinely distributed pipeline in which information accumulates redundantly across the team and no agent acts as a single point of failure.

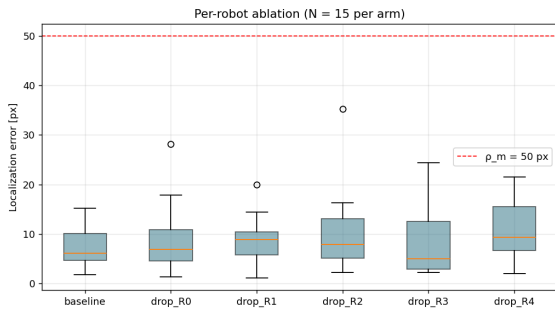


Fig. 5: Per-robot ablation boxplot. All six conditions sit within a narrow band, with all medians below $\rho_m = 50$ px.

H. Threats to validity

All experiments use the 900×600 map and the advection–diffusion plume; the analytical Pasquill–Gifford model is implemented but was not used here, and would likely amplify the downwind bias in $\bar{e}_x$. Sample sizes are modest for the contrast experiments ($N = 30$ for Lloyd vs. Random Walk and $N = 10$ per arm for Mode 2), below the level at which $\sim 20\%$ effects can be confidently resolved — this is directly visible in the reported standard deviations relative to the mean differences. The robots are point masses with velocity-limited integrators; realistic ground-vehicle dynamics would slow the four time intervals proportionally but should not affect the localization error, which is determined by the perception stack rather than the motion model.

VII. CONCLUSIONS

This work implements a distributed multi-robot hazard localization system that extends the consensus-based approach of Facinelli et al. [1]. The main methodological contribution is replacing the spiral localization heuristic with a Distributed Kalman Filter in information form, which gives principled uncertainty estimates while keeping the system fully decentralised. Around this core we added an active-patrol Voronoi coverage strategy, a scout-alert mechanism that compensates

for poor connectivity, and a trust-weighted gossip variant that excludes drifting agents. Results are reported as mean $\pm$ standard deviation over 365 paired-seed simulations across five experiments.

The headline empirical findings are: (i) 100/100 baseline runs inside the $\rho_m = 50$ px tolerance and 99/100 inside 25 px at $r_c = 500$; (ii) 15/15 completion at $r_c = 200$ px even though the communication graph is disconnected 100% of the time, evidence that local gossip inside transient geographic clusters substitutes for global algebraic connectivity; (iii) connectivity buys speed ($4.5\times$ faster from $r_c = 200$ to $r_c = 500$) much more than accuracy (the error is essentially flat, $9.5 \rightarrow 7.3$ px, over the same range); (iv) graceful fault tolerance — 10/10 runs inside tolerance with two of five robots degraded, in line with the $k < n/2$ bound of Pasqualetti–Bicchi–Bullo [6]; (v) leave-one-out ablation shows no critical agent (mean error varies by only 2.5 px across the five drop conditions, less than the baseline standard deviation). Lloyd–Voronoi outperforms Random Walk in point estimates (chiefly $\approx 30\%$ shorter patrol), but the spread of the paired differences (≈ 8 px around a -0.6 px mean) is too large for the contrast to be conclusive at $N = 30$.

a) Limitations: The system is restricted to 2D ground robots with first-order kinematics; wind dynamics are slow stochastic drift rather than a measured atmospheric series; multi-source localization is not supported by the current single-mode DKF (would require a mixture-of-Gaussians belief or parallel filters with source matching); and the Lloyd-vs.-Random-Walk comparison remains underpowered at $N = 30$.

b) Future Work: Three natural extensions: rerun all experiments under the Pasquill–Gifford plume model to see how much the downwind bias amplifies; extend to a heterogeneous team with UAV scouts using the framework of [10]; and replace the trust-EMA in (17) with the system-theoretic detection approach of [6] for stronger guarantees on the maximum tolerable number of faulty agents.

REFERENCES

- [1] D. Facinelli, M. Larcher, D. Brunelli, and D. Fontanelli, “Cooperative UAVs gas monitoring using distributed consensus,” in *Proc. IEEE Annual Computer Software and Applications Conference (COMPSAC)*, 2019, pp. 463–468, arXiv:1907.07279.
- [2] R. Olfati-Saber, “Distributed Kalman filtering for sensor networks,” *Proc. IEEE Conf. on Decision and Control*, pp. 5492–5498, 2007.
- [3] J. Cortés, S. Martínez, T. Karatas, and F. Bullo, “Coverage control for mobile sensing networks,” *IEEE Trans. Robotics and Automation*, vol. 20, no. 2, pp. 243–255, 2004.
- [4] M. Mesbahi and M. Egerstedt, *Graph Theoretic Methods in Multiagent Networks*. Princeton University Press, 2010.
- [5] S. Boyd, A. Ghosh, B. Prabhakar, and D. Shah, “Randomized gossip algorithms,” *IEEE Trans. Information Theory*, vol. 52, no. 6, pp. 2508–2530, 2006.
- [6] F. Pasqualetti, A. Bicchi, and F. Bullo, “Consensus computation in unreliable networks: A system theoretic approach,” *IEEE Trans. Automatic Control*, vol. 57, no. 1, pp. 90–104, 2012.
- [7] R. Olfati-Saber, J. A. Fax, and R. M. Murray, “Consensus and cooperation in networked multi-agent systems,” *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, 2007.
- [8] H. Ishida, Y. Wada, and H. Matsukura, “Chemical sensing in robotic applications: A review,” *IEEE Sensors Journal*, vol. 12, no. 11, pp. 3163–3173, 2012, reviews plume-following localization; explains the downwind bias of peak-tracking strategies.

- [9] Z. C. Lipton and J. Steinhardt, “Troubling trends in machine learning scholarship,” *arXiv preprint arXiv:1807.03341*, 2018, discussion of ablation studies and the practice of leave-one-out evaluation to justify claims of necessity.
- [10] M. Andreetto, M. Pacher, D. Macii, L. Palopoli, and D. Fontanelli, “A distributed strategy for target tracking and rendezvous using UAVs relying on visual information only,” vol. 7, no. 10, 2018.